

Deccan Education Society's
Fergusson College (Autonomous), Pune
Department of Computer Science

A

**Project Report
on**

“Fake news Detection using ML and LLM”

In partial fulfillment of Post Graduate course

in

M.Sc. Computer Science – I

(Semester -II)

CSC-560 Project

SUBMITTED BY

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Fergusson College (Autonomous), Pune
Department of Computer Science**

CERTIFICATE

This is to certify that the project entitled
Fake news Detection using ML and LLM submitted by

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in partial fulfillment of the requirement of the completion of M.Sc. (C.S)-I [Semester-II],
has been carried out by them under our guidance satisfactorily during the academic year
2025-2026.

Place: Pune

Date: / /2026

Dr. Kavita Khobragade
Head,
Department of Computer Science
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Project Guide: Prof.Arti Nimgaonkar

Examiners Name

Sign

1. _____

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1. Introduction

With the rapid growth of digital media and social networking platforms, the spread of misinformation and fake news has become a serious problem. Fake news can mislead people, influence public opinion, and create social unrest. Detecting fake news manually is difficult because of the large amount of content published daily on the internet.

Artificial Intelligence and Machine Learning techniques can help analyze news articles and determine whether the information is reliable or misleading. In this project, an AI-powered fake news detection system is developed using Large Language Models (LLM) and web technologies.

The system allows users to:

- **Paste news text and check if it is real or fake**
- **Paste news article URLs and verify their authenticity**
- **View trending news headlines**
- **View history of previously analyzed news and URLs**

The system uses Flask for backend development, HTML/CSS for the user interface, and LLM-based analysis for credibility verification.

1.1 Existing System

Currently, many social media platforms and news websites allow the rapid sharing of information. However, there are several limitations in the existing system:

- 1. Fake news spreads quickly through social media platforms.**
- 2. Users usually verify news manually by searching multiple sources.**
- 3. Many fake news detection systems rely only on traditional machine learning models.**
- 4. Most systems cannot analyze live news URLs directly.**
- 5. Few systems provide real-time AI reasoning using large language models.**

Because of these limitations, detecting misinformation becomes difficult for general users.

1.2 Need of the System

The need for an automated fake news detection system arises due to the following reasons:

- **Increasing spread of misinformation on the internet.**
- **Lack of reliable tools for verifying news authenticity.**
- **Need for fast and automated verification of news articles.**
- **Requirement for AI-based reasoning to explain why news is fake or real.**

The proposed system helps users quickly verify news credibility using AI technology.

1.3 Overview of the Project

The AI Fake News Detection System is a web-based application that analyzes news content using Large Language Models (LLM).

The system includes the following main modules:

- 1. News Text Analysis**
Users can paste news content, and the system analyzes it using an LLM to determine if it is fake or real.
- 2. URL News Analysis**
Users can paste a news article URL. The system extracts the article text using web scraping and verifies it using the AI model.
- 3. Trending News Section**
Displays current trending headlines from online news sources.
- 4. History Tracking**
Stores previously analyzed news text and URLs and displays them in the history section.
- 5. LLM-Based Analysis**
The system uses an LLM to evaluate credibility and provide explanations.

This system helps users quickly determine whether news content is trustworthy.

2. Analysis

2.1 Feasibility Study

Feasibility analysis determines whether the project is practical and achievable.

2.1.1 Technical Feasibility

The project is technically feasible because:

- Python provides powerful libraries for AI and web development.**
- Flask allows rapid backend development.**
- Large Language Models provide advanced reasoning capabilities.**
- Web scraping can extract article content from URLs.**

Technologies used are widely available and easy to integrate.

2.1.2 Economical Feasibility

The system is economically feasible because:

- It uses open-source technologies such as Python and Flask.
- Development tools are free.
- Deployment cost is minimal.

Therefore, the project can be implemented at low cost.

2.1.3 Operational Feasibility

The system is easy to operate because:

- It has a simple web interface.
- Users only need to paste text or a URL.
- Results are displayed instantly.

This makes the system practical for general users.

2.2 Hardware and Software Requirements

Hardware Requirements

- **Processor:** Intel i3 or higher
- **RAM:** Minimum 4 GB
- **Storage:** 10 GB free space
- **Internet connection**

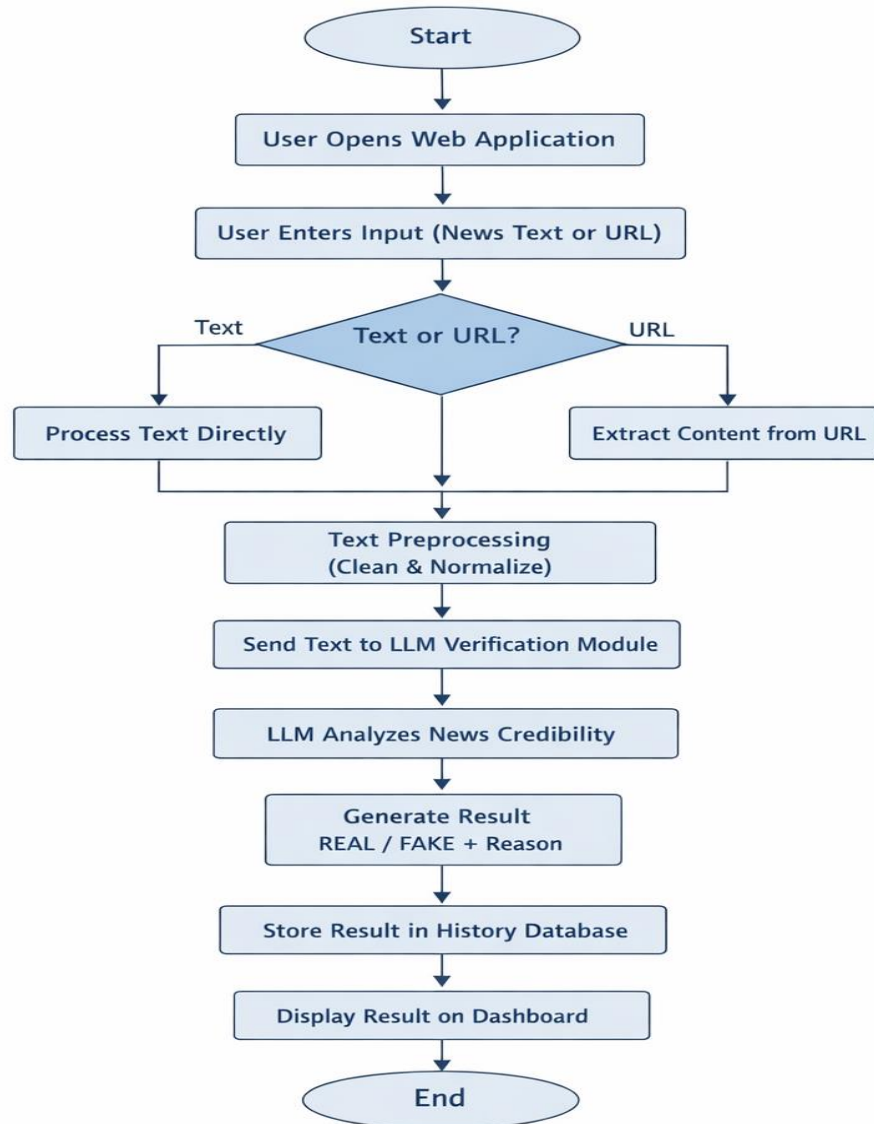
Software Requirements

- **Operating System:** Windows / Linux / MacOS
- **Programming Language:** Python
- **Framework:** Flask
- **Frontend:** HTML, CSS, JavaScript
- **Libraries:**
 - Flask
 - Requests
 - BeautifulSoup
 - LLM API integration
- **Browser:** Chrome / Edge / Firefox

3. Design

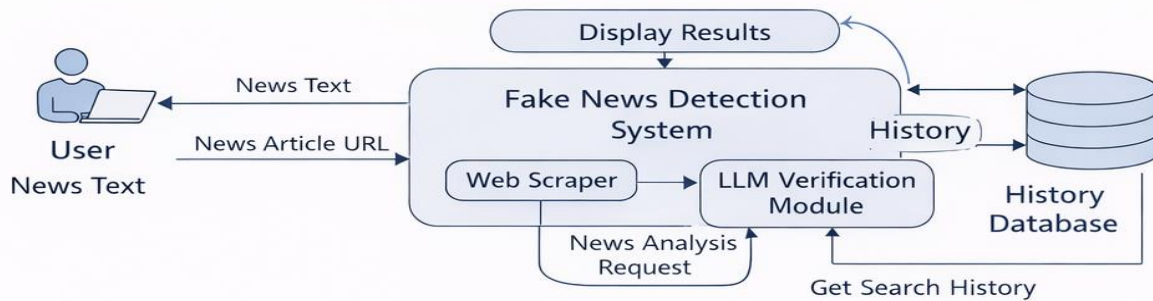
3.1 Algorithm

AI Fake News Detection System

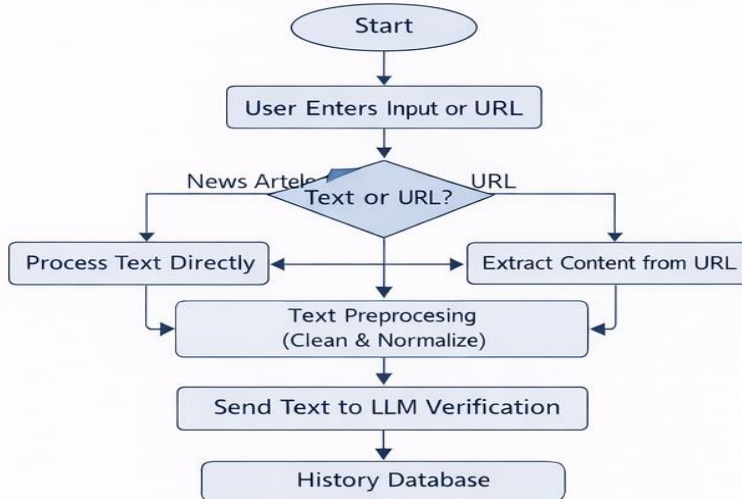


3.2 Software Engineering Diagrams

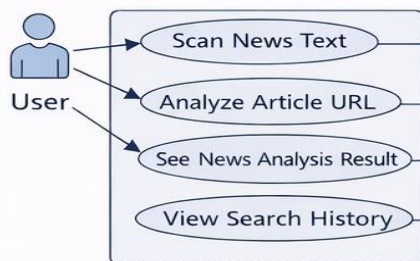
Data Flow Diagram (DFD)



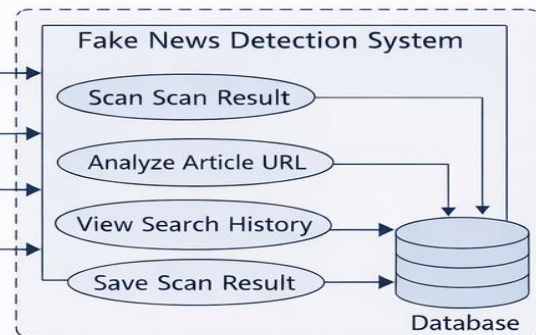
Activity Diagram



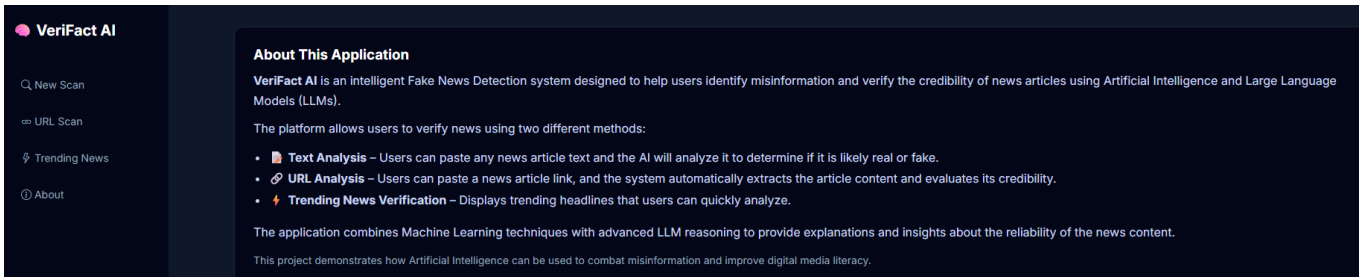
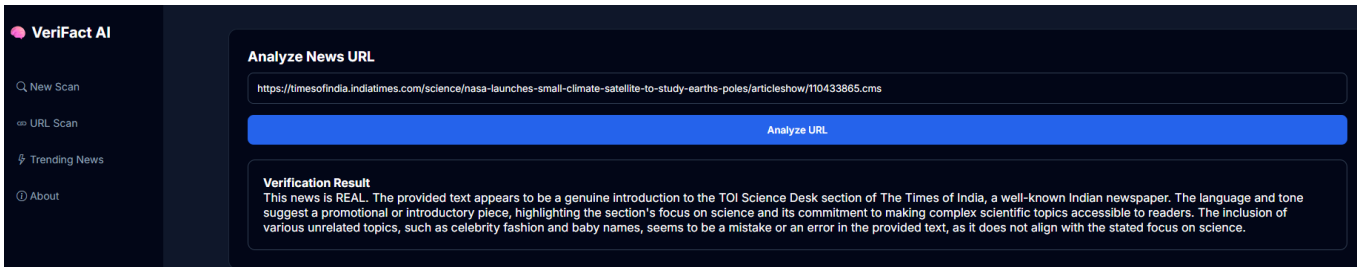
Use Case Diagram



Use Case Diagram



3.3 Input / Output Screens



4. Testing

4.1 Importance of Testing

Testing ensures that the system works correctly and produces accurate results. It helps identify errors and improve system reliability.

4.2 Types of Testing

Unit Testing

Testing individual modules such as:

- URL extraction
- LLM verification

Integration Testing

Ensures that frontend, backend, and AI modules work together properly.

System Testing

Testing the complete system with real news articles.

4.3 Test Cases

Test Case	Input	Expected Output
1	Real news article	Result = Real
2	Fake news headline	Result = Fake
3	Valid URL	Article extracted and analyzed
4	Invalid URL	Error message

5. Reports

The system generates the following reports:

- Fake news analysis results
- AI explanation
- Trending news list
- History of analyzed news and URLs

These reports help users track previously analyzed content.

6.Drawbacks and Limitations

Despite its advantages, the system has some limitations:

- Accuracy depends on the LLM response.
- Some websites may block scraping.
- History data is stored temporarily in memory.
- Internet connection is required.

7.Future Enhancements and Conclusion

Future Enhancements

The system can be improved with the following features:

- Persistent database storage for history
- Integration with fact-checking APIs
- Real-time news verification
- AI confidence score
- User login system
-

Conclusion

The AI Fake News Detection System provides a simple and efficient solution for identifying misinformation online. By combining web scraping, LLM analysis, and a user-friendly interface, the system enables users to quickly verify the credibility of news articles.

This project demonstrates the potential of Artificial Intelligence in combating misinformation and promoting reliable information on the internet.

8. Bibliography

1. Python Documentation
2. Flask Documentation
3. BeautifulSoup Documentation
4. Research papers on Fake News Detection
5. OpenAI / LLM documentation